

1. Overview

This document provides guidance about methods and approaches to achieve de-identification in accordance with the Health Insurance Portability and Accountability Act of 1996 (HIPAA) Privacy Rule. The guidance explains and answers questions regarding the two methods that can be used to satisfy the Privacy Rule's de-identification standard: Expert Determination and Safe Harbor¹. This guidance is intended to assist covered entities to understand what is de-identification, the general process by which de-identified information is created, and the options available for performing de-identification.

In developing this guidance, the Office for Civil Rights (OCR) solicited input from stakeholders with practical, technical and policy experience in de-identification. OCR convened stakeholders at a workshop consisting of multiple panel sessions held March 8-9, 2010, in Washington, DC. Each panel addressed a specific topic related to the Privacy Rule's de-identification methodologies and policies. The workshop was open to the public and each panel was followed by a question and answer period. More information about the workshop, including a summary, can be found at <http://www.hhs.gov/ocr/privacy/hipaa/understanding/coveredentities/De-identification/deidentificationworkshop2010.html>. A webcast of the workshop can be viewed through streaming video from the website.

1.1. Protected Health Information

The HIPAA Privacy Rule protects most "individually identifiable health information" held or transmitted by a covered entity or its business associate, in any form or medium, whether electronic, on paper, or oral. The Privacy Rule calls this information *protected health information* (PHI).² Protected health information is information, including demographic information, which relates to:

- the individual's past, present, or future physical or mental health or condition,
- the provision of health care to the individual, or
- the past, present, or future payment for the provision of health care to the individual,

and that identifies the individual or for which there is a reasonable basis to believe can be used to identify the individual. Protected health information includes many common identifiers (e.g., name, address, birth date, Social Security Number) when they can be associated with the health information listed above.

¹ The Health Information Technology for Economic and Clinical Health (HITECH) Act was enacted as part of the American Recovery and Reinvestment Act of 2009 (ARRA). Section 13424(c) of the HITECH Act requires the Secretary of HHS to issue guidance on how best to implement the requirements for the de-identification of health information contained in the Privacy Rule.

² Protected health information (PHI) is defined as individually identifiable health information transmitted or maintained by a covered entity or its business associates in any form or medium (45 CFR 160.103). The definition exempts a small number of categories of individually identifiable health information, such as individually identifiable health information found in employment records held by a covered entity in its role as an employer.

For example, a medical record, laboratory report, or hospital bill would be PHI because each document would contain a patient's name and/or other identifying information associated with the health data content.

By contrast, a health plan report that only noted the average age of health plan members was 45 years would not be PHI because that information, although developed by aggregating information from individual plan member records, does not identify any individual plan members and there is no reasonable basis to believe that it could be used to identify an individual.

The relationship with health information is fundamental. Identifying information alone, such as personal names, residential addresses, or phone numbers, would not necessarily be designated as PHI. For instance, if such information was reported as part of a publicly accessible data source, such as a phone book, then this information would not be PHI because it is not related to health data (see above). If such information was listed with health condition, health care provision or payment data, such as an indication that the individual was treated at a certain clinic, then this information would be PHI.

1.2. Covered Entities, Business Associates, and PHI

In general, the protections of the Privacy Rule apply to information held by covered entities and their business associates. HIPAA defines a covered entity as 1) a health care provider that conducts certain standard administrative and financial transactions in electronic form; 2) a health care clearinghouse; or 3) a health plan.³ A business associate is a person or entity (other than a member of the covered entity's workforce) that performs certain functions or activities on behalf of, or provides certain services to, a covered entity that involve the use or disclosure of protected health information. A covered entity may use a business associate to de-identify PHI on its behalf only to the extent such activity is authorized by their business associate agreement.

See the OCR website <http://www.hhs.gov/ocr/privacy/> for detailed information about the Privacy Rule and how it protects the privacy of health information.

1.3. De-identification and its Rationale

The increasing adoption of health information technologies in the United States accelerates their potential to facilitate beneficial studies that combine large, complex data sets from multiple sources. The process of de-identification, by which identifiers are removed from the health information, mitigates privacy risks to individuals and thereby supports the secondary use of data for comparative effectiveness studies, policy assessment, life sciences research, and other endeavors.

³ Detailed definitions and explanations of these covered entities and their varying types can be found in the "Covered Entity Charts" available through the OCR website, at <http://www.hhs.gov/ocr/privacy/hipaa/understanding/coveredentities/index.html>. Discussion of business associates can be found at <http://www.hhs.gov/ocr/privacy/hipaa/understanding/coveredentities/businessassociates.html>

The Privacy Rule was designed to protect individually identifiable health information through permitting only certain uses and disclosures of PHI provided by the Rule, or as authorized by the individual subject of the information. However, in recognition of the potential utility of health information even when it is not individually identifiable, §164.502(d) of the Privacy Rule permits a covered entity or its business associate to create information that is not individually identifiable by following the de-identification standard and implementation specifications in §164.514(a)-(b). These provisions allow the entity to use and disclose information that neither identifies nor provides a reasonable basis to identify an individual.⁴ As discussed below, the Privacy Rule provides two de-identification methods: 1) a formal determination by a qualified expert; or 2) the removal of specified individual identifiers as well as absence of actual knowledge by the covered entity that the remaining information could be used alone or in combination with other information to identify the individual.

Both methods, even when properly applied, yield de-identified data that retains some risk of identification. Although the risk is very small, it is not zero, and there is a possibility that de-identified data could be linked back to the identity of the patient to which it corresponds.

Regardless of the method by which de-identification is achieved, the Privacy Rule does not restrict the use or disclosure of de-identified health information, as it is no longer considered protected health information.

1.4. The De-identification Standard

Section 164.514(a) of the HIPAA Privacy Rule provides the standard for de-identification of protected health information. Under this standard, health information is not individually identifiable if it does not identify an individual and if the covered entity has no reasonable basis to believe it can be used to identify an individual.

§ 164.514 Other requirements relating to uses and disclosures of protected health information.

(a) *Standard: de-identification of protected health information.* Health information that does not identify an individual and with respect to which there is no reasonable basis to believe that the information can be used to identify an individual is not individually identifiable health information.

Sections 164.514(b) and (c) of the Privacy Rule contain the implementation specifications that a covered entity must follow to meet the de-identification standard. As summarized in Figure 1, the Privacy Rule provides two methods by which health information can be designated as de-identified.

⁴ In some instances, other federal protections also may apply, such as those found in Family Educational Rights and Privacy Act (FERPA) or the Common Rule.

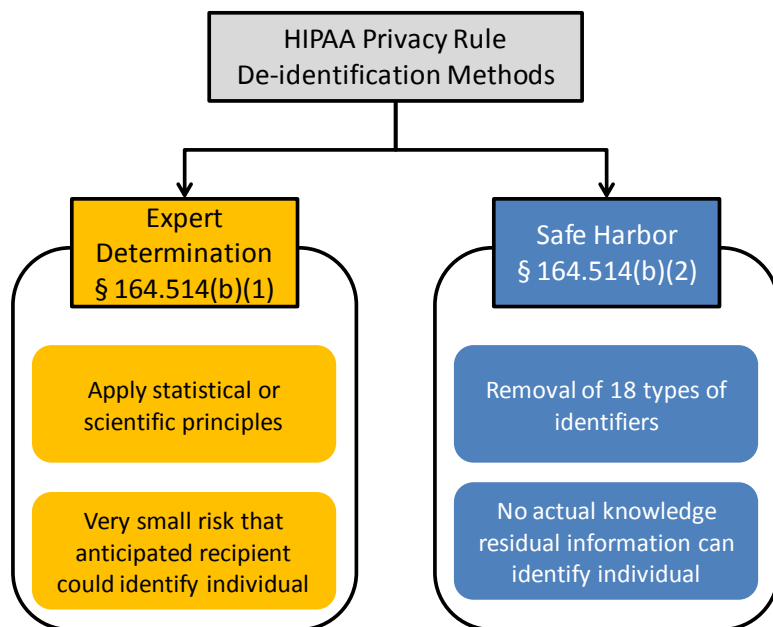


Figure 1. Two methods to achieve de-identification in accordance with the HIPAA Privacy Rule.

The first is the “Expert Determination” method:

(b) *Implementation specifications: requirements for de-identification of protected health information.* A covered entity may determine that health information is not individually identifiable health information only if:

- (1) A person with appropriate knowledge of and experience with generally accepted statistical and scientific principles and methods for rendering information not individually identifiable:
 - (i) Applying such principles and methods, determines that the risk is very small that the information could be used, alone or in combination with other reasonably available information, by an anticipated recipient to identify an individual who is a subject of the information; and
 - (ii) Documents the methods and results of the analysis that justify such determination; or

The second is the “Safe Harbor” method:

(2)(i) The following identifiers of the individual or of relatives, employers, or household members of the individual, are removed:

- | |
|---|
| (A) Names |
| (B) All geographic subdivisions smaller than a state, including street address, city, county, precinct, ZIP code, and their equivalent geocodes, except for the initial three digits of the ZIP code if, according to the current publicly available data from the Bureau of the Census: <ol style="list-style-type: none"> (1) The geographic unit formed by combining all ZIP codes with the same three initial digits contains more than 20,000 people; and (2) The initial three digits of a ZIP code for all such geographic units |

containing 20,000 or fewer people is changed to 000	
(C) All elements of dates (except year) for dates that are directly related to an individual, including birth date, admission date, discharge date, death date, and all ages over 89 and all elements of dates (including year) indicative of such age, except that such ages and elements may be aggregated into a single category of age 90 or older	
(D) Telephone numbers	(L) Vehicle identifiers and serial numbers, including license plate numbers
(E) Fax numbers	(M) Device identifiers and serial numbers
(F) Email addresses	(N) Web Universal Resource Locators (URLs)
(G) Social security numbers	(O) Internet Protocol (IP) addresses
(H) Medical record numbers	(P) Biometric identifiers, including finger and voice prints
(I) Health plan beneficiary numbers	(Q) Full-face photographs and any comparable images
(J) Account numbers	(R) Any other unique identifying number, characteristic, or code, except as permitted by paragraph (c) of this section [Paragraph (c) is presented below in the section “Re-identification”]; and
(K) Certificate/license numbers	

(ii) The covered entity does not have actual knowledge that the information could be used alone or in combination with other information to identify an individual who is a subject of the information.

Satisfying either method would demonstrate that a covered entity has met the standard in §164.514(a) above. De-identified health information created following these methods is no longer protected by the Privacy Rule because it does not fall within the definition of PHI. Of course, de-identification leads to information loss which may limit the usefulness of the resulting health information in certain circumstances. As described in the forthcoming sections, covered entities may wish to select de-identification strategies that minimize such loss.

Re-identification

The implementation specifications further provide direction with respect to *re-identification*, specifically the assignment of a unique code to the set of de-identified health information to permit re-identification by the covered entity.

(c) *Implementation specifications: re-identification.* A covered entity may assign a code or other means of record identification to allow information de-identified under this section to be re-identified by the covered entity, provided that:

- (1) *Derivation.* The code or other means of record identification is not derived from or related to information about the individual and is not otherwise capable of being translated so as to identify the individual; and
- (2) *Security.* The covered entity does not use or disclose the code or other means of record identification for any other purpose, and does not disclose the mechanism for re-identification.

If a covered entity or business associate successfully undertook an effort to identify the subject of de-identified information it maintained, the health information now related to a specific individual would again be protected by the Privacy Rule, as it would meet the definition of PHI. Disclosure of a code or other means of record identification designed to enable coded or otherwise de-identified information to be re-identified is also considered a disclosure of PHI.

1.5. Preparation for De-identification

The importance of documentation for which values in health data correspond to PHI, as well as the systems that manage PHI, for the de-identification process cannot be overstated. Esoteric notation, such as acronyms whose meaning are known to only a select few employees of a covered entity, and incomplete description may lead those overseeing a de-identification procedure to unnecessarily redact information or to fail to redact when necessary. When sufficient documentation is provided, it is straightforward to redact the appropriate fields. See section 3.10 for a more complete discussion.

In the following two sections, we address questions regarding the Expert Determination method (Section 2) and the Safe Harbor method (Section 3).

3. Guidance on Satisfying the Safe Harbor Method

In §164.514(b), the Safe Harbor method for de-identification is defined as follows:

(2)(i) The following identifiers of the individual or of relatives, employers, or household members of the individual, are removed:

(A) Names	
(B) All geographic subdivisions smaller than a state, including street address, city, county, precinct, ZIP code, and their equivalent geocodes, except for the initial three digits of the ZIP code if, according to the current publicly available data from the Bureau of the Census: (1) The geographic unit formed by combining all ZIP codes with the same three initial digits contains more than 20,000 people; and (2) The initial three digits of a ZIP code for all such geographic units containing 20,000 or fewer people is changed to 000	
(C) All elements of dates (except year) for dates that are directly related to an individual, including birth date, admission date, discharge date, death date, and all ages over 89 and all elements of dates (including year) indicative of such age, except that such ages and elements may be aggregated into a single category of age 90 or older	
(D) Telephone numbers	(L) Vehicle identifiers and serial numbers, including license plate numbers
(E) Fax numbers	(M) Device identifiers and serial numbers
(F) Email addresses	(N) Web Universal Resource Locators (URLs)
(G) Social security numbers	(O) Internet Protocol (IP) addresses
(H) Medical record numbers	(P) Biometric identifiers, including finger and voice prints
(I) Health plan beneficiary numbers	(Q) Full-face photographs and any comparable images
(J) Account numbers	(R) Any other unique identifying number, characteristic, or code, except as permitted by paragraph (c) of this section; and
(K) Certificate/license numbers	

(ii) The covered entity does not have actual knowledge that the information could be used alone or in combination with other information to identify an individual who is a subject of the information.

3.1. When can ZIP codes be included in de-identified information?

Covered entities may include the first three digits of the ZIP code if, according to the current publicly available data from the Bureau of the Census: (1) The geographic unit formed by combining all ZIP codes with the same three initial digits contains more than 20,000 people; or (2) the initial three digits of a ZIP code for all such geographic units

containing 20,000 or fewer people is changed to 000. This means that the initial three digits of ZIP codes may be included in de-identified information *except* when the ZIP codes contain the initial three digits listed in the Table below. In those cases, the first three digits must be listed as 000.

OCR published a final rule on August 14, 2002, that modified certain standards in the Privacy Rule. The preamble to this final rule identified the initial three digits of ZIP codes, or *ZIP code tabulation areas (ZCTAs)*, that must change to 000 for release. 67 FR 53182, 53233-53234 (Aug. 14, 2002)).

Utilizing 2000 Census data, the following three-digit ZCTAs have a population of 20,000 or fewer persons. To produce a de-identified data set utilizing the safe harbor method, all records with three-digit ZIP codes corresponding to these three-digit ZCTAs must have the ZIP code changed to 000. *Covered entities should not, however, rely upon this listing or the one found in the August 14, 2002 regulation if more current data has been published.*

The 17 restricted ZIP codes are:

036	692	878
059	790	879
063	821	884
102	823	890
203	830	893
556	831	

The Department notes that these three-digit ZIP codes are based on the five-digit ZIP Code Tabulation Areas created by the Census Bureau for the 2000 Census. This new methodology also is briefly described below, as it will likely be of interest to all users of data tabulated by ZIP code. The Census Bureau will not be producing data files containing U.S. Postal Service ZIP codes either as part of the Census 2000 product series or as a post Census 2000 product. However, due to the public's interest in having statistics tabulated by ZIP code, the Census Bureau has created a new statistical area called the Zip Code Tabulation Area (ZCTA) for Census 2000. The ZCTAs were designed to overcome the operational difficulties of creating a well-defined ZIP code area by using Census blocks (and the addresses found in them) as the basis for the ZCTAs. In the past, there has been no correlation between ZIP codes and Census Bureau geography. Zip codes can cross State, place, county, census tract, block group, and census block boundaries. The geographic designations the Census Bureau uses to tabulate data are relatively stable over time. For instance, census tracts are only defined every ten years. In contrast, ZIP codes can change more frequently. Because of the ill-defined nature of ZIP code boundaries, the Census Bureau has no file (crosswalk) showing the relationship between US Census Bureau geography and U.S. Postal Service ZIP codes.

ZCTAs are generalized area representations of U.S. Postal Service (USPS) ZIP code service areas. Simply put, each one is built by aggregating the Census 2000 blocks, whose addresses use a given ZIP code, into a ZCTA which gets that ZIP code assigned as its ZCTA code. They represent the majority USPS five-digit ZIP code found in a given area. For those areas where it is difficult to determine the prevailing five-digit ZIP code, the higher-level three-digit ZIP code is used for the ZCTA code. For further information, go to: <http://www.census.gov/geo/www/gazetteer/places2k.html>.

The Bureau of the Census provides information regarding population density in the United States. Covered entities are expected to rely on the most current publicly available Bureau of Census data regarding ZIP codes. This information can be downloaded from, or queried at, the American Fact Finder website (<http://factfinder.census.gov>). As of the publication of this guidance, the information can be extracted from the detailed tables of the “Census 2000 Summary File 1 (SF 1) 100-Percent Data” files under the “Decennial Census” section of the website. The information is derived from the Decennial Census and was last updated in 2000. It is expected that the Census Bureau will make data available from the 2010 Decennial Census in the near future. This guidance will be updated when the Census makes new information available.

3.2. May parts or derivatives of any of the listed identifiers be disclosed consistent with the Safe Harbor Method?

No. For example, a data set that contained patient initials, or the last four digits of a Social Security number, would not meet the requirement of the Safe Harbor method for de-identification.

3.3. What are examples of dates that are not permitted according to the Safe Harbor Method?

Elements of dates that are not permitted for disclosure include the day, month, and any other information that is more specific than the year of an event. For instance, the date “January 1, 2009” could not be reported at this level of detail. However, it could be reported in a de-identified data set as “2009”.

Many records contain dates of service or other events that imply age. Ages that are explicitly stated, or implied, as over 89 years old must be recoded as 90 or above. For example, if the patient’s year of birth is 1910 and the year of healthcare service is reported as 2010, then in the de-identified data set the year of birth should be reported as “on or before 1920.” Otherwise, a recipient of the data set would learn that the age of the patient is approximately 100.

3.4. Can dates associated with test measures for a patient be reported in accordance with Safe Harbor?

No. Dates associated with test measures, such as those derived from a laboratory report, are directly related to a specific individual and relate to the provision of health care. Such dates are protected health information. As a result, no element of a date (except as described in 3.3. above) may be reported to adhere to Safe Harbor.

3. 5. What constitutes “any other unique identifying number, characteristic, or code” with respect to the Safe Harbor method of the Privacy Rule?

This category corresponds to any unique features that are not explicitly enumerated in the Safe Harbor list (A-Q), but could be used to identify a particular individual. Thus, a covered entity must ensure that a data set stripped of the explicitly enumerated identifiers also does not contain any of these unique features. The following are examples of such features:

Identifying Number

There are many potential identifying numbers. For example, the preamble to the Privacy Rule at 65 FR 82462, 82712 (Dec. 28, 2000) noted that “Clinical trial record numbers are included in the general category of ‘any other unique identifying number, characteristic, or code.’”

Identifying Code

A code corresponds to a value that is derived from a non-secure encoding mechanism. For instance, a code derived from a secure hash function without a secret key (e.g., “salt”) would be considered an identifying element. This is because the resulting value would be susceptible to compromise by the recipient of such data. As another example, an increasing quantity of electronic medical record and electronic prescribing systems assign and embed barcodes into patient records and their medications. These barcodes are often designed to be unique for each patient, or event in a patient’s record, and thus can be easily applied for tracking purposes. See the discussion of re-identification.

Identifying Characteristic

A *characteristic* may be anything that distinguishes an individual and allows for identification. For example, a unique identifying characteristic could be the occupation of a patient, if it was listed in a record as “current President of State University.”

Many questions have been received regarding what constitutes “any other unique identifying number, characteristic or code” in the Safe Harbor approach, §164.514(b)(2)(i)(R), above. Generally, a code or other means of record identification that is derived from PHI would have to be removed from data de-identified following the safe harbor method. To clarify what must be removed under (R), the implementation specifications at §164.514(c) provide an exception with respect to “re-identification” by the covered entity. The objective of the paragraph is to permit covered entities to assign certain types of codes or other record identification to the de-identified information so that it may be re-identified by the covered entity at some later date. Such codes or other means of record identification assigned by the covered entity are not considered direct identifiers that must be removed under (R) if the covered entity follows the directions provided in §164.514(c).

3.6. What is “actual knowledge” that the remaining information could be used either alone or in combination with other information to identify an individual who is a subject of the information?

In the context of the Safe Harbor method, actual knowledge means clear and direct knowledge that the remaining information could be used, either alone or in combination with other information, to identify an individual who is a subject of the information. This means that a covered entity has actual knowledge if it concludes that the remaining information could be used to identify the individual. The covered entity, in other words, is aware that the information is not actually de-identified information.

The following examples illustrate when a covered entity would fail to meet the “actual knowledge” provision.

Example 1: Revealing Occupation

Imagine a covered entity was aware that the occupation of a patient was listed in a record as “former president of the State University.” This information in combination with almost any additional data – like age or state of residence – would clearly lead to an identification of the patient. In this example, a covered entity would not satisfy the de-identification standard by simply removing the enumerated identifiers in §164.514(b)(2)(i) because the risk of identification is of a nature and degree that a covered entity must have concluded that the information could identify the patient. Therefore, the data would not have satisfied the de-identification standard’s Safe Harbor method unless the covered entity made a sufficient good faith effort to remove the “occupation” field from the patient record.

Example 2: Clear Familial Relation

Imagine a covered entity was aware that the anticipated recipient, a researcher who is an employee of the covered entity, had a family member in the data (e.g., spouse, parent, child, or sibling). In addition, the covered entity was aware that the data would provide sufficient context for the employee to recognize the relative. For instance, the details of a complicated series of procedures, such as a primary surgery followed by a set of follow-up surgeries and examinations, for a person of a certain age and gender, might permit the recipient to comprehend that the data pertains to his or her relative’s case. In this situation, the risk of identification is of a nature and degree that the covered entity must have concluded that the recipient could clearly and directly identify the individual in the data. Therefore, the data would not have satisfied the de-identification standard’s Safe Harbor method.

Example 3: Publicized Clinical Event

Rare clinical events may facilitate identification in a clear and direct manner. For instance, imagine the information in a patient record revealed that a patient gave birth to an unusually large number of children at the same time. During the year of this event, it is highly possible that this occurred for only one individual in the hospital (and perhaps the country). As a result, the event was reported in the popular media, and the covered entity was aware of this media exposure. In this case, the risk of identification is of a nature and degree that the covered entity must have concluded that the individual subject of the information could be

identified by a recipient of the data. Therefore, the data would not have satisfied the de-identification standard's Safe Harbor method.

Example 4: Knowledge of a Recipient's Ability

Imagine a covered entity was told that the anticipated recipient of the data has a table or algorithm that can be used to identify the information, or a readily available mechanism to determine a patient's identity. In this situation, the covered entity has actual knowledge because it was informed outright that the recipient can identify a patient, unless it subsequently received information confirming that the recipient does not in fact have a means to identify a patient. Therefore, the data would not have satisfied the de-identification standard's Safe Harbor method.

3.7. If a covered entity knows of specific studies about methods to re-identify health information or use de-identified health information alone or in combination with other information to identify an individual, does this necessarily mean a covered entity has *actual knowledge* under the Safe Harbor method?

No. Much has been written about the capabilities of researchers with certain analytic and quantitative capacities to combine information in particular ways to identify health information.^{32,33,34,35} A covered entity may be aware of studies about methods to identify remaining information or using de-identified information alone or in combination with other information to identify an individual. However, a covered entity's mere knowledge of these studies and methods, by itself, does not mean it has "actual knowledge" that these methods would be used with the data it is disclosing. OCR does not expect a covered entity to presume such capacities of all potential recipients of de-identified data. This would not be consistent with the intent of the Safe Harbor method, which was to provide covered entities with a simple method to determine if the information is adequately de-identified.

3. 8. Must a covered entity suppress all personal names, such as physician names, from health information for it to be designated as de-identified?

No. Only names of the individuals associated with the corresponding health information (i.e., the subjects of the records) and of their relatives, employers, and household members must be suppressed. There is no explicit requirement to remove the names of

³² K. El Emam, F. Dankar, R. Vaillancourt, T. Roffey, and M. Lysyk. Evaluating the risk of re-identification of patients from hospital prescription records. *Canadian Journal of Hospital Pharmacy*. 2009; 62(4): 307-319.

³³ G. Loukides, J. Denny, and B. Malin. The disclosure of diagnosis codes can breach research participants privacy. *Journal of the American Medical Informatics Association Annual Symposium*. 2010; 17(3): 322-327.

³⁴ B. Malin and L. Sweeney. How (not) to protect genomic data privacy in a distributed network: using trail re-identification to evaluate and design anonymity protection systems. *Journal of Biomedical Informatics*. 2004; 37(3): 179-192.

³⁵ L. Sweeney. Data sharing under HIPAA: 12 years later. A presentation at the Workshop on the HIPAA Privacy Rule's De-Identification Standard. Washington, DC. March 8-9, 2010.

providers or workforce members of the covered entity or business associate. At the same time, there is also no requirement to retain such information in a de-identified data set.

Beyond the removal of names related to the patient, the covered entity would need to consider whether additional personal names contained in the data should be suppressed to meet the *actual knowledge* specification. Additionally, other laws or confidentiality concerns may support the suppression of this information.

3.9. Must a covered entity use a data use agreement when sharing de-identified data to satisfy the Safe Harbor Method?

No. The Privacy Rule does not limit how a covered entity may disclose information that has been de-identified. However, nothing prevents a covered entity from asking a recipient of de-identified information to enter into a data use agreement, such as is required for release of a *limited data set* under the Privacy Rule. This agreement may prohibit re-identification. Of course, the use of a data use agreement does not substitute for any of the specific requirements of the Safe Harbor method. Further information about data use agreements can be found on the OCR website.³⁶ Covered entities may make their own assessments whether such additional oversight is appropriate.

3.10. Must a covered entity remove protected health information from free text fields to satisfy the Safe Harbor Method?

PHI may exist in different types of data in a multitude of forms and formats in a covered entity. This data may reside in highly structured database tables, such as billing records. Yet, it may also be stored in a wide range of documents with less structure and written in natural language, such as discharge summaries, progress notes, and laboratory test interpretations. These documents may vary with respect to the consistency and the format employed by the covered entity.

The de-identification standard makes no distinction between data entered into standardized fields and information entered as free text (i.e., structured and unstructured text) -- an identifier listed in the Safe Harbor standard must be removed regardless of its location in a record if it is recognizable as an identifier.

Whether additional information must be removed falls under the *actual knowledge* provision; the extent to which the covered entity has actual knowledge that residual information could be used to individually identify a patient. Clinical narratives in which a physician documents the history and/or lifestyle of a patient are information rich and may provide context that readily allows for patient identification.

Medical records are comprised of a wide range of structured and unstructured (also known as “free text”) documents. In structured documents, it is relatively clear which fields contain the identifiers that must be removed following the Safe Harbor method.

³⁶ Supra note 28.

For instance, it is simple to discern when a feature is a name or a Social Security Number, provided that the fields are appropriately labeled. However, many researchers have observed that identifiers in medical information are not always clearly labeled.^{37,38} As such, in some electronic health record systems it may be difficult to discern what a particular term or phrase corresponds to (e.g., is 5/97 a date or a ratio?). It also is important to document when fields are derived from the Safe Harbor listed identifiers. For instance, if a field corresponds to the first initials of names, then this derivation should be noted. De-identification is more efficient and effective when data managers explicitly document when a feature or value pertains to identifiers. Health Level 7 (HL7) and the International Standards Organization (ISO) publish best practices in documentation and standards that covered entities may consult in this process.

Example Scenario 1

The free text field of a patient's medical record notes that the patient is the Executive Vice President of the state university. The covered entity must remove this information.

Example Scenario 2

The intake notes for a new patient include the stand-alone notation, "Newark, NJ." It is not clear whether this relates to the patient's address, the location of the patient's previous health care provider, the location of the patient's recent auto collision, or some other point. The phrase may be retained in the data.

³⁷ D. Dorr, W. Phillips, S. Phansalkar, S. Sims, and J. Hurdle. Assessing the difficulty and time cost of de-identification in clinical narratives. *Methods of Information in Medicine*. 2006; 45(3): 246-252.

³⁸ O. Uzuner, Y. Luo, and P. Szolovits. Evaluating the state-of-the-art in automatic de-identification. *Journal of the American Medical Informatics Association*. 2007; 14(5): 550-563.